

```
!pip install fastapi uvicorn gradio beautifulsoup4 requests nest_asyncio
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Requirement already satisfied: fastapi in /usr/local/lib/python3.12/dist-packages (0.109.1)
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```

```
import requests
from bs4 import BeautifulSoup

def crawl_quotes():
    url = "http://quotes.toscrape.com/"
    res = requests.get(url)
    soup = BeautifulSoup(res.text, "html.parser")

    data = []
    for q in soup.select(".quote")[1:20]:
        text = q.select_one(".text").get_text()
        author = q.select_one(".author").get_text()
        tags = [t.get_text() for t in q.select(".tag")]

        data.append((text, author, ", ".join(tags)))

    return data
```

```
import sqlite3

conn = sqlite3.connect("quotes.db")
cur = conn.cursor()

cur.execute("""
CREATE TABLE IF NOT EXISTS quotes (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    text TEXT,
    author TEXT,
    tags TEXT
)
""")

data = crawl_quotes()

cur.execute("DELETE FROM quotes") # 초기화
for d in data:
    cur.execute("INSERT INTO quotes (text, author, tags) VALUES (?, ?, ?)", d)

conn.commit()
conn.close()

print("저장 완료:", len(data))
```

저장 완료: 10

```
from fastapi import FastAPI
import sqlite3
import random

app = FastAPI()

@app.get("/quotes")
def get_quotes():
    conn = sqlite3.connect("quotes.db")
    cur = conn.cursor()
    cur.execute("SELECT * FROM quotes")
    data = cur.fetchall()
    conn.close()
    return data

@app.get("/random")
def random_quote():
    conn = sqlite3.connect("quotes.db")
    cur = conn.cursor()
    cur.execute("SELECT * FROM quotes")
    data = cur.fetchall()
    conn.close()
    return random.choice(data)
```

```
import gradio as gr
import requests

def get_data():
    return requests.get("http://localhost:8000/quotes").json()

def get_random():
    return requests.get("http://localhost:8000/random").json()

with gr.Blocks() as demo:
    gr.Markdown("# 📖 Quotes Dashboard")

    btn1 = gr.Button("전체 조회")
    btn2 = gr.Button("랜덤 명언")

    out = gr.JSON()

    btn1.click(get_data, outputs=out)
    btn2.click(get_random, outputs=out)
```

```
from gradio import mount_gradio_app
app = mount_gradio_app(app, demo, path="/")
```

new /

```

import nest_asyncio
nest_asyncio.apply()

from pyngrok import ngrok
import uvicorn
import threading
import time

# 기존 터널 제거
ngrok.kill()

# 토큰 입력
from getpass import getpass
token = getpass("ngrok 토큰 입력: ")
ngrok.set_auth_token(token)

# 서버 실행
def run():
    uvicorn.run(app, host="0.0.0.0", port=8000)

threading.Thread(target=run).start()

# 서버 준비 시간
time.sleep(3)

# ngrok 연결
public_url = ngrok.connect(8000)
print("🔥 ngrok URL:", public_url)

```

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ngrok 토큰 입력: . . . . .
INFO:      Started server process [13446]
INFO:      Waiting for application startup.
INFO:      Application startup complete.
ERROR:    [Errno 98] error while attempting to bind on address ('0.0.0.0', 8000): address already in use
INFO:      Waiting for application shutdown.
INFO:      Application shutdown complete.
🔥 ngrok URL: NgrokTunnel: "https://pungent-gentile-hydrogen.ngrok-free.dev" -> "http://localhost:8000"

```

이 밑에 URL 을 새창에서 실행

<https://pungent-gentile-hydrogen.ngrok-free.dev/docs>

<https://pungent-gentile-hydrogen.ngrok-free.dev/gradio>

토큰 : 3DLkLAprOS0Nvz1nbA8JmwSm8n1_2W1LEUgATFFVrTMbS9ucc